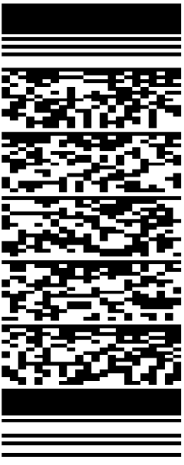


Company: Equinor		Well: 31/5-7	Field: Eos	Rig Name: West Hercules	Country: Norway	
VISION Resistivity Dual Frequency Recorded Mode, Run 9 Schlumberger						
Company: Equinor		Well: 31/5-7				
Field: Eos		Rig Name: West Hercules				
Country: Norway		Latitude: 60° 34' 35.14" N				
Longitude: 3° 26' 36.13" E		UWID:		Rig Name: West Hercules		
Block: 31/5		Rig Type: 31/5		Semi-Submersible		
FL: Norwegian North Sea		FL1: Section: 8 1/2 in.				
FL2: Job no.: 19SCA0085						
Log Measured From: - Drill Floor: 31.00 m Permanent Datum: - Mean Sea Level Ground Level: 307.00 m Other Services: Direction & Inclination (Surveys) Annular Pressure While Drilling geoVISION Bit Resistivity						
Acquisition Dates: 29-Dec-2019 -- 30-Dec-2019		Log Interval: 2544.00(m) -- 2593.00(m)				
Index Types: Measured Depth		Index Scales: 1:200				
Depth Source: Driller's Depth		Depth Sensor: 3rd Party Depth				
Print Type: Final		Spud Date: 02-Dec-2019				
						

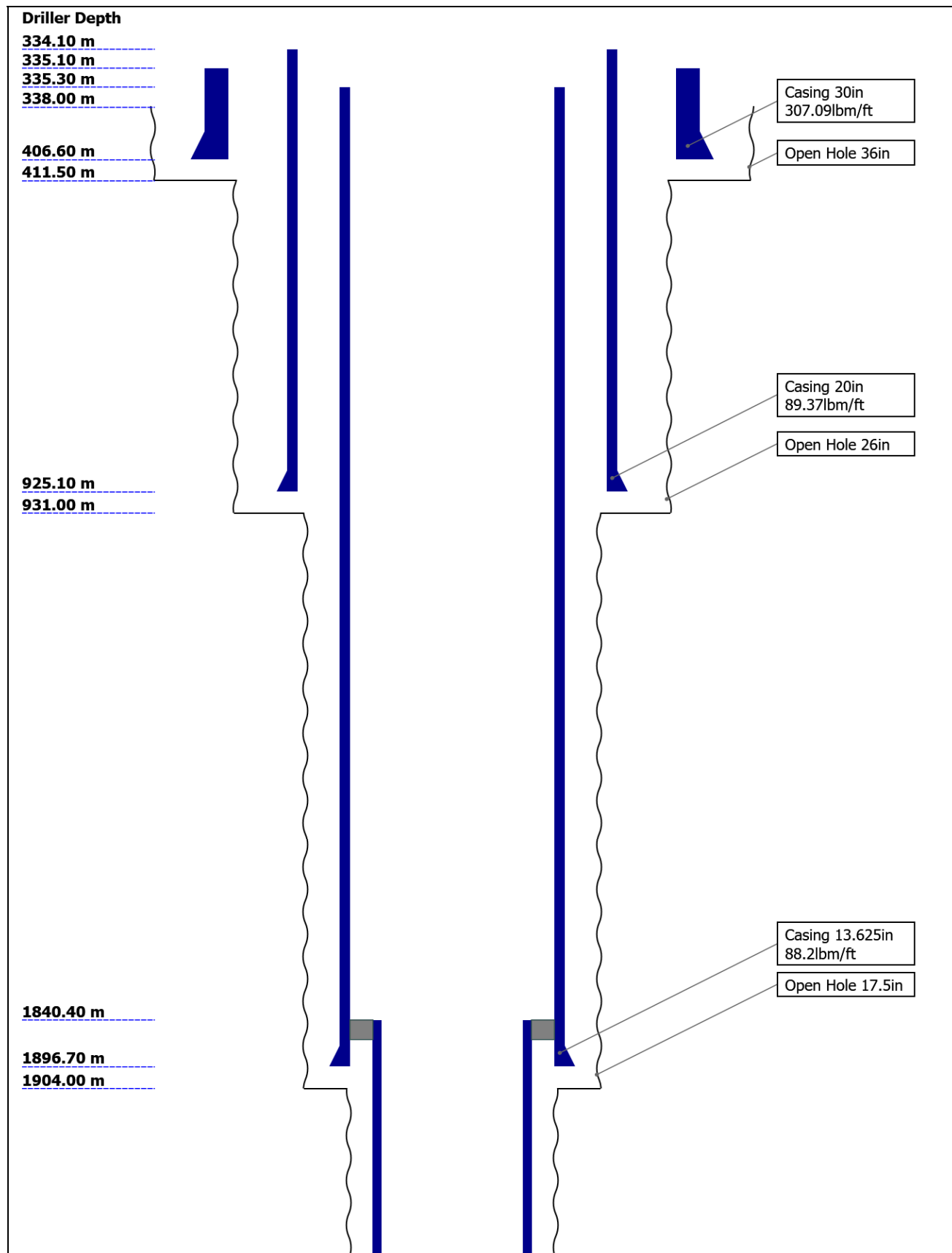
Disclaimer

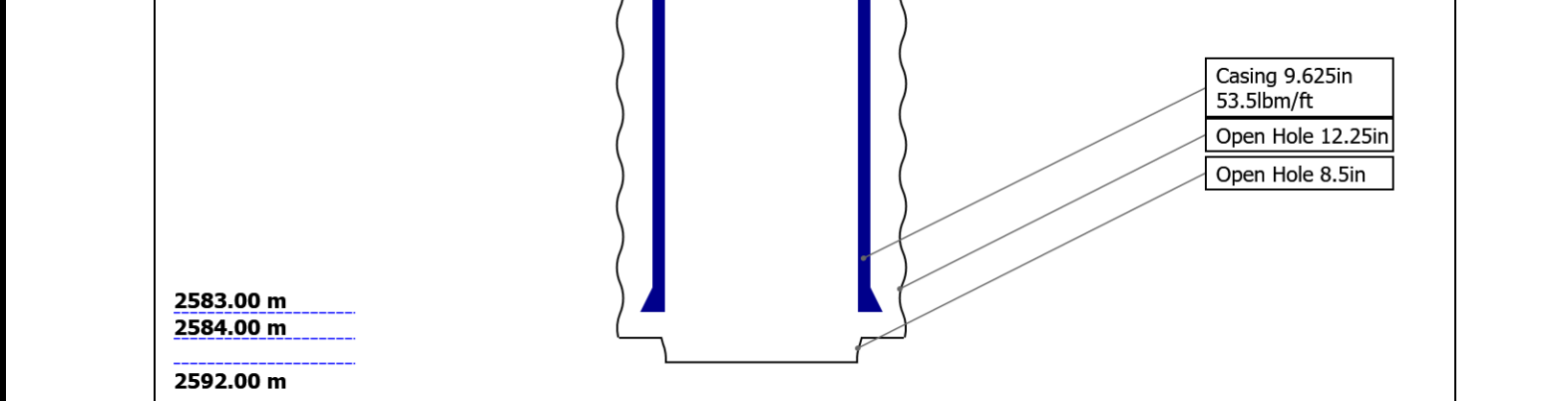
THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

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Well Sketch





Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25	8.5	
Top Driller (m)	338	411.5	931	1904	2584	
Bottom Driller (m)	411.5	931	1904	2584	2592	
Casing						
Size (in)	30	20	13.625	9.625		
Weight (lbm/ft)	307.09	89.37	88.2	53.5		
Inner Diameter (in)	27.067	18.73	12.375	8.535		
Grade	X56	N80	P110	P110		
Top Driller (m)	335.1	334.1	335.3	1840.4		
Bottom Driller (m)	406.6	925.1	1896.7	2583		

Operational Run Summary

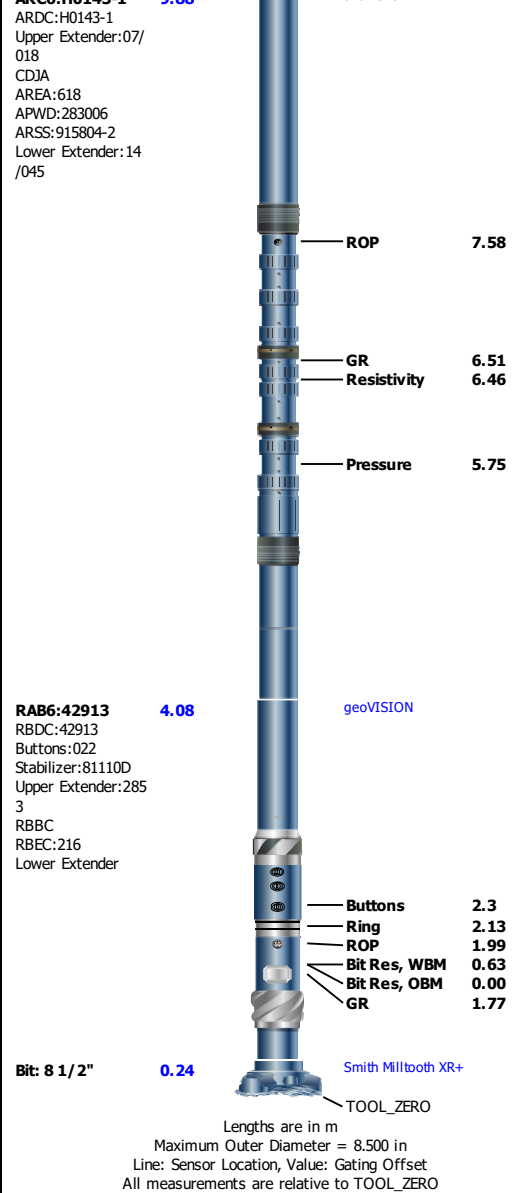
Parameter (unit)	Run 9					
Date Log Started	29-Dec-2019					
Time Log Started	06:12:03					
Date Log Finished	30-Dec-2019					
Time Log Finished	05:04:18					
Bit Size (in)	8.500					
Bit Start Depth (m)	2584.00					
Bit Stop Depth (m)	2592.00					
Top Log Interval (m)	2584.70					
Bottom Log Interval (m)	2591.22					
Max Hole Deviation (deg)	1.10					
Azimuth of Max Deviation (deg)	133.27					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	S.Kroevel					
Witnessed By	J.Eikenes					
Service Order Number	19SCA0085					

Borehole Fluids

Parameter(unit)	Run 9					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	38					
Source of Sample	Active Tank					
Salinity (ppm)	93984					
Density (g/cm3)	1.15					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.07 @ 18					
RMF @ Meas Temp (ohm.m@degC)	0.06 @ 18					
RMC @ Meas Temp (ohm.m@degC)	0.1 @ 20					
RM @ BHT (ohm.m@degC)	0.05 @ 38					
RMF @ BHT (ohm.m@degC)	0.04 @ 38					
RMC @ BHT (ohm.m@degC)	0.07 @ 38					
Total Solid (%)						
High Gravity Solids (%)						

Remarks and Equipment Summary

Run 9: Toolstring				Run 9: Remarks	
Equip name TELE675:E4388 Length 18.34 MP name TeleScope Offset MSSU675:30848 Upper Extender MDC675:E4388 MMA:2507 MDI:3666 PMVC:236 PMEA:803 MTA:2162 MTK675:004 MSSD675:1005482-1 Lower Extender, Logging 	All depths referenced to driller's depth. See EOWR for depth tracking details.				
	All data from tool memory. All data acquired while drilling.				
	Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.				
	Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.				
	Run 9 POOH at 2592 m to perform coring.				



Run 9

VISION Resistivity - Dual Frequency

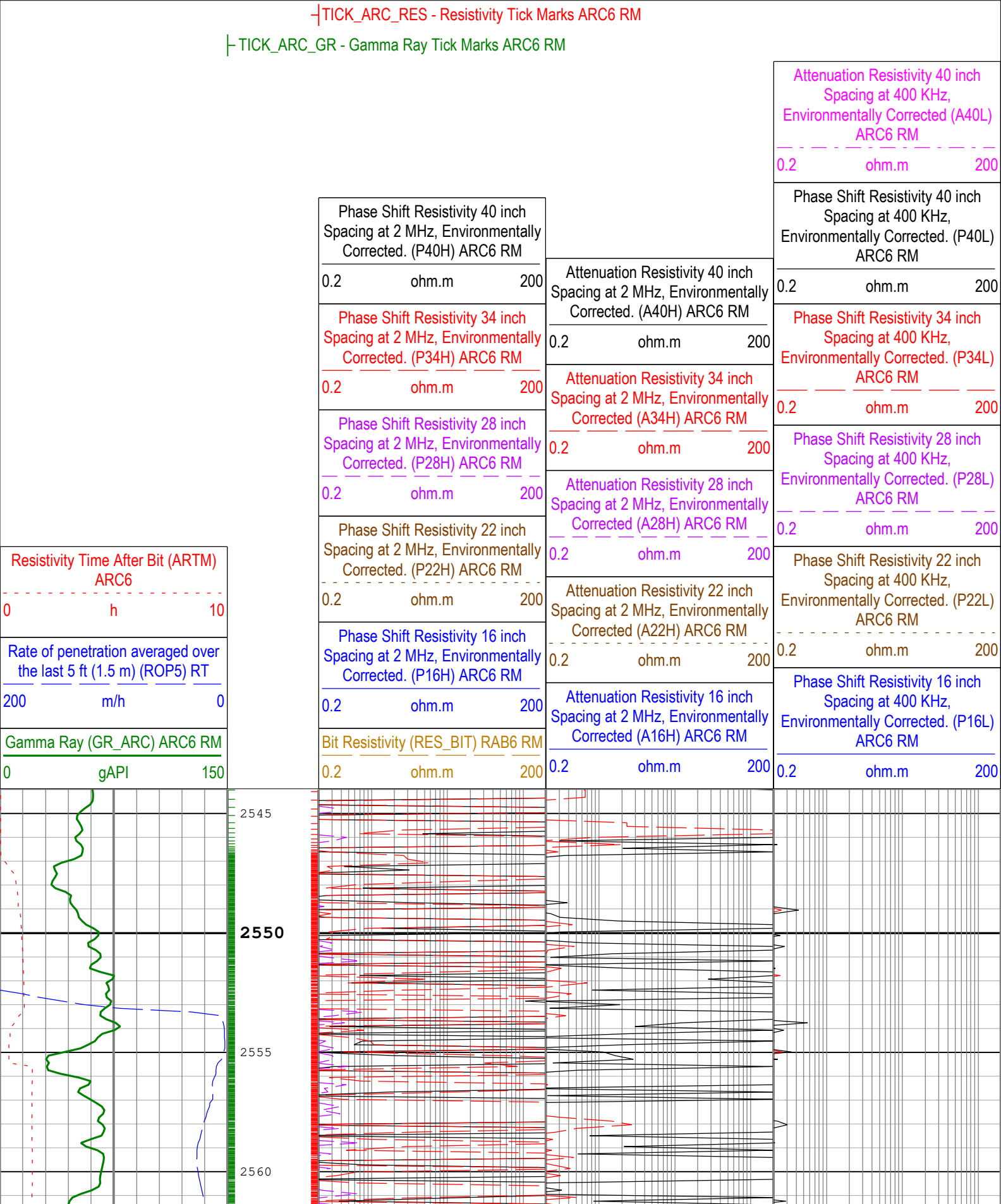
Software Version

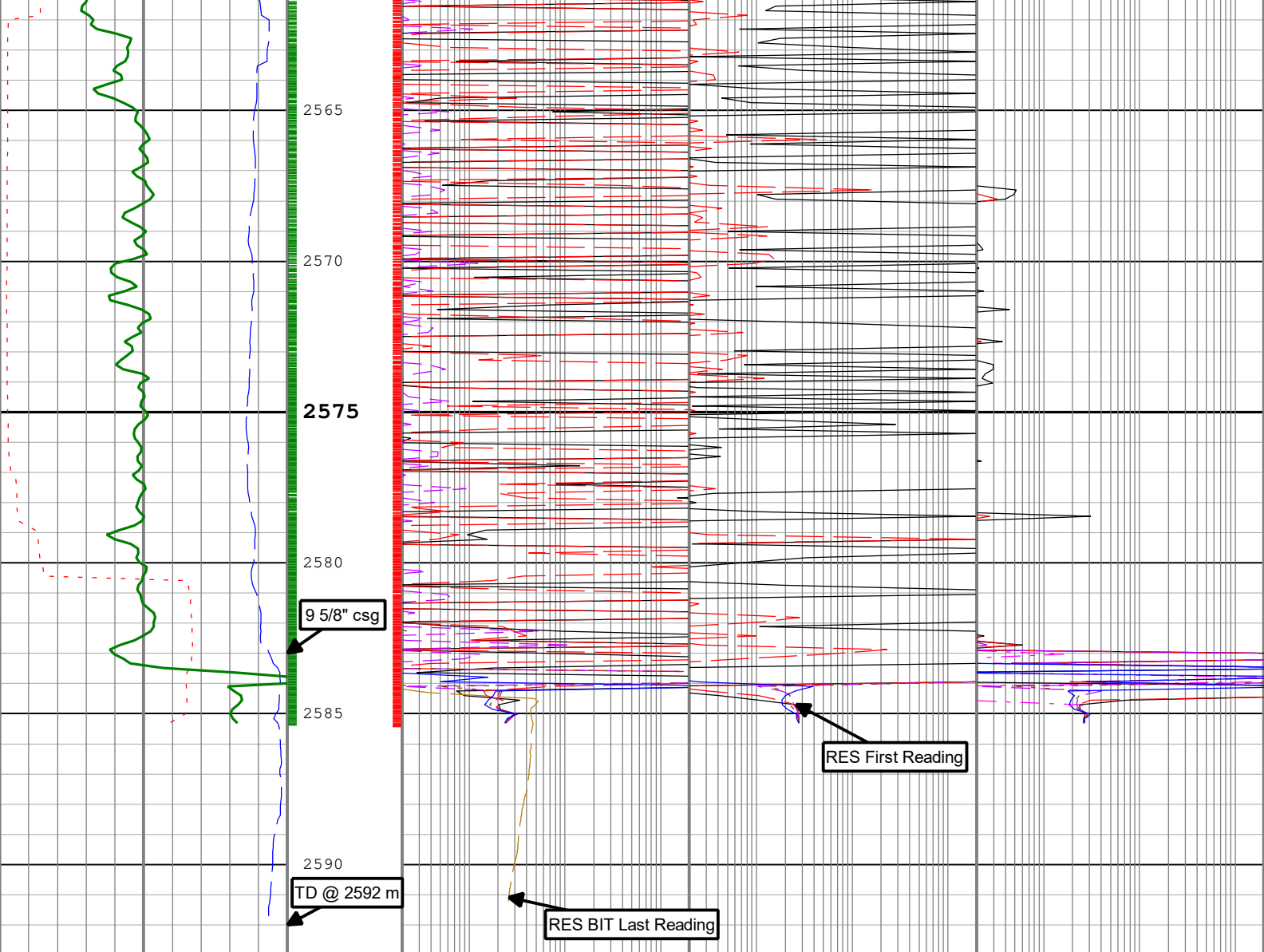
Acquisition System		Version	
Maxwell 2019.2		9.2.113335.3100	
Application Patch		DnM_TestKit-XceedCPU41-2019.2_9.2.114181	
Computation	Description	Version	
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family	9.2.113335.3100	
ARC6GammaRay	ARC6 Gamma Ray Computation Package for both Real-time and Recorded Mode	9.2.113335.3100	
GVRRes	GVR Resistivity Computation Package for both Real-time and Recorded Mode	9.2.113335.3100	
Tool Interface	System Version	Loaded Version	
HSPM	2019.0c.001	9.2.113335.3100	
Tool Elements	Description	Software Version	Firmware Version
RBEC	Electronics Chassis Assembly for RAB6-C	9.2.113335.3100	V8.8B
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 6.75 Inch Tool Drill Collar	9.2.113335.3100	V9.7B

Pass Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 9	Drilling	Down	720.04 m	2591.89 m	29-Dec-2019 6:12:03 AM	30-Dec-2019 5:04:18 AM	Yes

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 20-Jan-2020 10:51:11





Resistivity Time After Bit (ARTM) ARC6	Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC6 RM	Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC6 RM	Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC6 RM
0h10	0.2ohm.m200	0.2ohm.m200	0.2ohm.m200
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC6 RM	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC6 RM	Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC6 RM
200m/h0	0.2ohm.m200	0.2ohm.m200	0.2ohm.m200
Gamma Ray (GR_ARC) ARC6 RM	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC6 RM	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC6 RM	Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC6 RM
0gAPI150	0.2ohm.m200	0.2ohm.m200	0.2ohm.m200
	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC6 RM	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC6 RM	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC6 RM
	0.2ohm.m200	0.2ohm.m200	0.2ohm.m200
	Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC6 RM	Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC6 RM	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC6 RM
	0.2ohm.m200	0.2ohm.m200	0.2ohm.m200
	Bit Resistivity (RES_BIT) RAB6 RM		
	0.2ohm.m200		

	0.2	ohm.m	200
Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC6 RM			
	0.2	ohm.m	200

└─TICK_ARC_GR - Gamma Ray Tick Marks ARC6 RM

└─TICK_ARC_RES - Resistivity Tick Marks ARC6 RM

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 20-Jan-2020 10:51:11

Channel Processing Parameters

Run 9: Parameters

Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC6	NO_TX_FAILED	
BH_COMPUTE	Borehole Effect Computation Option	ARC6	No	
BHK	Drilling Fluid Potassium Concentration	Borehole	6.7	%
BHT	Bottom Hole Temperature	Borehole	38	degC
BS	Bit Size	DNMSESSION	Depth Zoned	in
DEPTH_SEL	Depth Selection Parameter	DNMSESSION	Driller's Depth	
DFD	Drilling Fluid Density	Borehole	1.15	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
ERRCT	Percentage Error Cutoff	ARC6	4.5	%
GRSE_RM	Generalized Mud Resistivity Selection for Recorded Mode	Borehole	REMS(RM)	
GRSH	Gamma Ray Shale	ARC6	1000	gAPI
GTSE_RM	Generalized Temperature Selection for Recorded Mode	Borehole	DHAT(RM)	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC6	2	ohm.m
INVAS_COMPUTE	Invasion Computation Option	ARC6	No	
LOW_BLEND	Low Resistivity Threshold for Blending	ARC6	1	ohm.m
MST	Mud Sample Temperature	Borehole	18	degC
MSWS	ARCWizard Model Switch Window Size	ARC6	152.4	cm
MULTIEFFECT_COMPUTE	Multi-effect Computation Option	ARC6	No	
PRTD	ARCWizard Preferred Resistivty Log for Rt Display while Multi-Effects	ARC6	P34B	
RMS	Resistivity of Mud Sample	Borehole	0.07	ohm.m
T1WM	ARCWizard Weight Multiplier to Measurements for Transmitter 1	ARC6	1	
T2WM	ARCWizard Weight Multiplier to Measurements for Transmitter 2	ARC6	1	
T3WM	ARCWizard Weight Multiplier to Measurements for Transmitter 3	ARC6	1	
T4WM	ARCWizard Weight Multiplier to Measurements for Transmitter 4	ARC6	1	
T5WM	ARCWizard Weight Multiplier to Measurements for Transmitter 5	ARC6	1	
ATMP_ARC	ARC Temperature Selection	ARC6	Annular	
TEMP_SEL_RAB	RAB Temperature Selection	RAB6	Tool	
UNIFORM_COMPUTE	Uniform Rock Computation Option	ARC6	No	

Depth Zone Parameters

Parameter	Value	Start (m)	Stop (m)
BS	12.25	2544	2584
BS	8.5	2584	2592

All depth are actual.

Tool Control Parameters

Run 9: Parameters

Parameter	Description	Tool	Value	Unit
OFFBTM_TH	Threshold for deciding whether the bit is off bottom	DNMSESSION	0	m

Calibration Report

RAB6 (GeoVision Resistivity 675) Calibration - Run 9

Primary Equipment :

Electronics Chassis

RBEC

216

M21V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 2 at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

M22V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 2 at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250	

M01V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 0 at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9950	1.0250	

M02V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 0 at T2 Calibration Coefficient		Master	1.0000	0.9750	0.9980	1.0250	

R1V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Ring at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

R2V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Ring at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250	

BDM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Deep at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0070	1.0250	

BDM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Deep at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0100	1.0250	

BMM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Medium at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9990	1.0250	

BMM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Medium at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250	

BSM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Shallow at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

BSM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Shallow at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250	

PGR - Gamma Ray: Blanket

Master (Time Frame File): 14:06:21 18-Dec-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.0000	0.7500	0.9838	1.2500	

ARC6 (Array Resistivity Compensated 675) Calibration - Run 9

Primary Equipment :

Elec. Chassis HP w/o AIM Receiver

AREA

618

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 15:26:50 07-Dec-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	7.951	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.997	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	4.584	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.904	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.142	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	1.927	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-1.871	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	1.878	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-1.879	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	1.856	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	7.970	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	6.989	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	4.593	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	4.889	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.160	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	-0.666	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	0.584	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	-0.629	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	0.582	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	-0.643	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:23:30 06-Jun-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.000	0.580	1.100	1.250	

Company: Equinor

Well: 31/5-7

Field: Eos
Rig Name: West Hercules
Country: Norway



Schlumberger

VISION Resistivity
Dual Frequency
Recorded Mode, Run 9